Skill-2(2520090018- M.Rasagnya)

STAGE 1 — Install tools

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [717 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [868 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [157 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [220 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [72.5 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [195 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [62.7 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [110 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [586 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [389 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [122 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [122 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [201 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [602 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [125 kB]
Get:21 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Fetched 5008 kB in 7s (681 kB/s)
194 packages can be upgraded. Run 'apt list --upgradable' to see them.
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd
tshark is already the newest version (4.6.4-1).
iperf3 is already the newest version (3.20-2).
mtr-tiny is already the newest version (0.95-1.1ubuntu2).
iproute2 is already the newest version (6.19.0-1ubuntu1.1).
iproute2 set to manually installed.
wireshark is already the newest version (4.6.4-1).
netcat-openbsd is already the newest version (1.234-1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 194
```

STAGE 2 — Build the network

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns add host1 sudo ip netns add rou
ter sudo ip netns add host2
[sudo: authenticate] Password:
Cannot create namespace file "/run/netns/host1": File exists
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns add host1 sudo ip netns add rou
ter sudo ip netns add host2
Cannot create namespace file "/run/netns/host1": File exists
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns add host1
Cannot create namespace file "/run/netns/host1": File exists
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns add router
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns add host1 sudo ip netns add rou
ter sudo ip netns add host2
Cannot create namespace file "/run/netns/host1": File exists
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns add host1 sudo ip netns add rou
ter sudo ip netns add host2
Cannot create namespace file "/run/netns/host1": File exists
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns add host1
Cannot create namespace file "/run/netns/host1": File exists
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns add router
Cannot create namespace file "/run/netns/router": File exists
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns add host2
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip link add veth-h1 type veth peer name
veth-r1sudo ip link add veth-r2 type veth peer name veth-h2
Error: either "dev" is duplicate, or "veth-r2" is garbage.
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip link add veth-h1 type veth peer name
veth-r1
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip link add veth-r2 type veth peer name
veth-h2
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip link set veth-h1 netns host1
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip link set veth-r1 netns router
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip link set veth-r2 netns router
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip link set veth-h2 netns host2
rasagnya_07@LAPTOP-7LMHP1NN:~$
```

STAGE 3 — Configure IP addresses

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host1 ip addr add 10.0.1.1
/30 dev veth-h1
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host1 ip link set veth-h1
up
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host1 ip link set lo up
```

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec router ip addr add 10.0.1.
2/30 dev veth-r1
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec router ip addr add 10.0.2.
1/30 dev veth-r2
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec router ip link set veth-r1
 up
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec router ip link set veth-r2
 up
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec router ip link set lo up
```

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host2 ip addr add 10.0.2.2
/30 dev veth-h2
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host2 ip link set veth-h2
up
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host2 ip link set lo up
```

STAGE 4 — Enable routing

```
rasagnya_07@LAPTOP-7LMHP1NN:~$  sudo ip netns exec router sysctl -w net.ipv4
.ip_forward=1
net.ipv4.ip_forward = 1
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host1 ip route add 10.0.2.
0/30 via 10.0.1.2
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host2 ip route add 10.0.1.
0/30 via 10.0.2.1
rasagnya_07@LAPTOP-7LMHP1NN:~$
```

STAGE 5 — Test connectivity

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.173 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.060 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.124 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2044ms
rtt min/avg/max/mdev = 0.060/0.119/0.173/0.046 ms
```

STAGE 6 — Add delay and loss

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec router tc qdisc add dev ve
th-r2 root netem delay 20ms loss 1%
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec router tc qdisc show dev v
eth-r2
qdisc netem 8001: root refcnt 17 limit 1000 delay 20ms loss 1% seed 91186375
85004924788
rasagnya_07@LAPTOP-7LMHP1NN:~$
```

STAGE 8 — Start three captures simultaneously

Terminal 1 — capture at host1

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.173 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.060 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.124 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2044ms
rtt min/avg/max/mdev = 0.060/0.119/0.173/0.046 ms
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec router tc qdisc add dev ve
th-r2 root netem delay 20ms loss 1%
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec router tc qdisc show dev v
eth-r2
qdisc netem 8001: root refcnt 17 limit 1000 delay 20ms loss 1% seed 91186375
85004924788
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host1 tshark -i veth-h1 -w
 /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
```

Terminal 2 — capture at router

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
```

Terminal 3 — capture at host2

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
```

STAGE 9 — Send real message from host1 to host2

STAGE 10 — What you see in real time

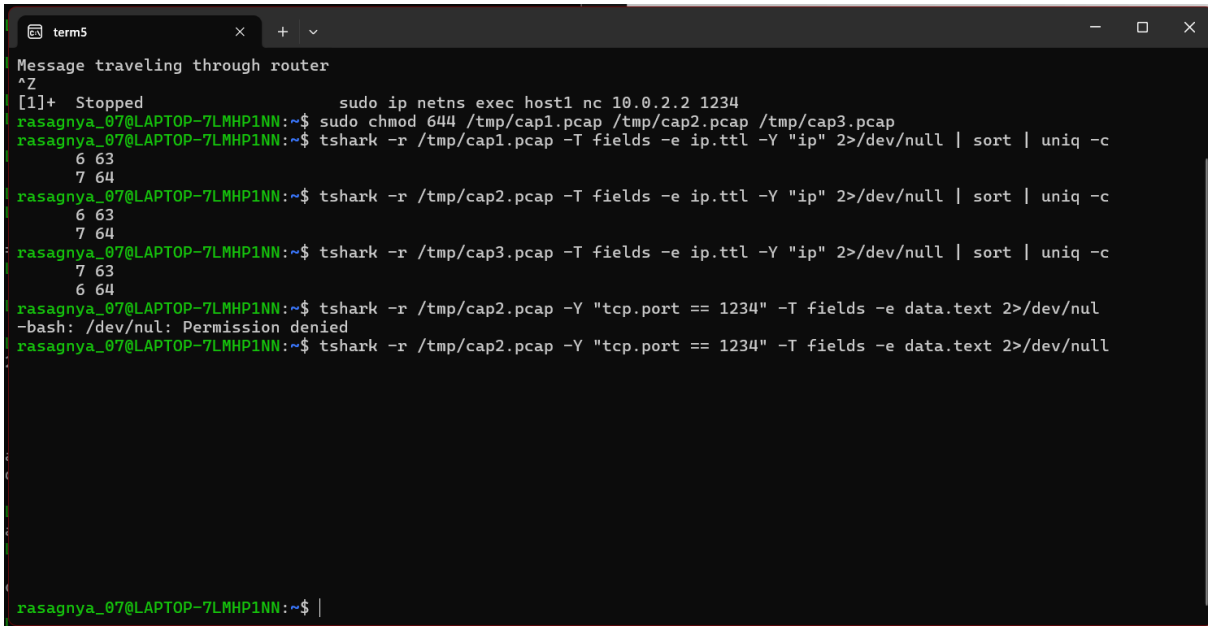
```
term4
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:
^Z
[1]+  Stopped                 sudo ip netns exec host2 nc -l 1234
rasagnya_07@LAPTOP-7LMHP1NN:~$
```

```
term5
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host1 nc 10.0.2.2 1234
[sudo: authenticate] Password:
Hello from host1!
This is Rasagnya
Computer Networks Lab
Three Point Capture Experiment
Message traveling through router
^Z
[1]+  Stopped                 sudo ip netns exec host1 nc 10.0.2.2 1234
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
rasagnya_07@LAPTOP-7LMHP1NN:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      6 63
      7 64
rasagnya_07@LAPTOP-7LMHP1NN:~$ tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      6 63
      7 64
rasagnya_07@LAPTOP-7LMHP1NN:~$ tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      7 63
      6 64
rasagnya_07@LAPTOP-7LMHP1NN:~$
```

Stage 11: Terminal 1, 2, 3 (tshark) show packet counts increasing:

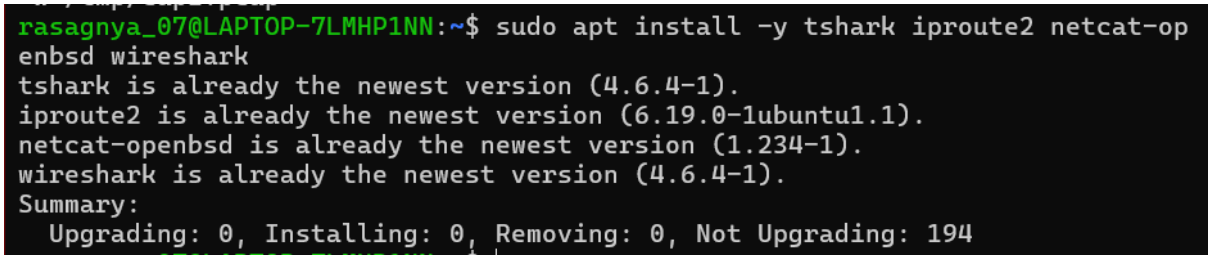
STAGE 12 — Fix permissions

STAGE 13 — Check TTL

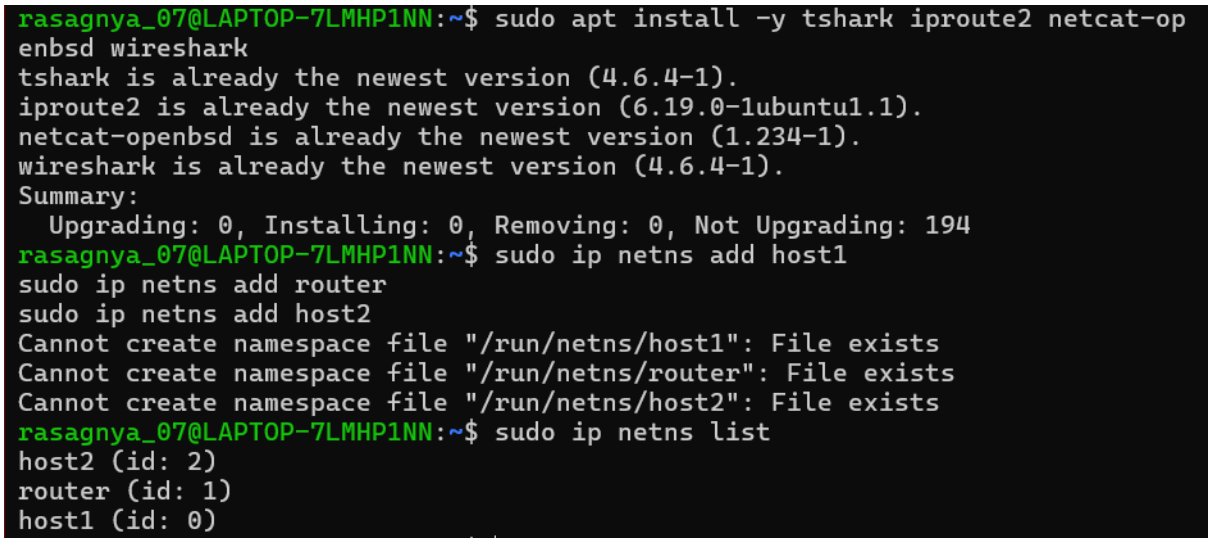


STAGE 14 — Read your actual message from the capture

Connect from host1:



Create 3 nodes:



Verify:

Create cables:

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
Cannot create namespace file "/run/netns/host1": File exists
Cannot create namespace file "/run/netns/router": File exists
Cannot create namespace file "/run/netns/host2": File exists
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns list
host2 (id: 2)
router (id: 1)
host1 (id: 0)
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip link add veth-h1 type veth peer name
veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
```

Plug cables into nodes:

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip link add veth-h1 type veth peer name
veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
```

Configure host1:

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip link add veth-h1 type veth peer name
veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host1 ip addr add 10.0.1.1
/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
Error: ipv4: Address already assigned.
```

Configure router:

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec router ip addr add 10.0.1.
2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
Error: ipv4: Address already assigned.
Error: ipv4: Address already assigned.
```

Configure host2:

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host2 ip addr add 10.0.2.2
/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
Error: ipv4: Address already assigned.
```

Enable routing:

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec router sysctl -w net.ipv4.
ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
RTNETLINK answers: File exists
RTNETLINK answers: File exists
```

Test connectivity:

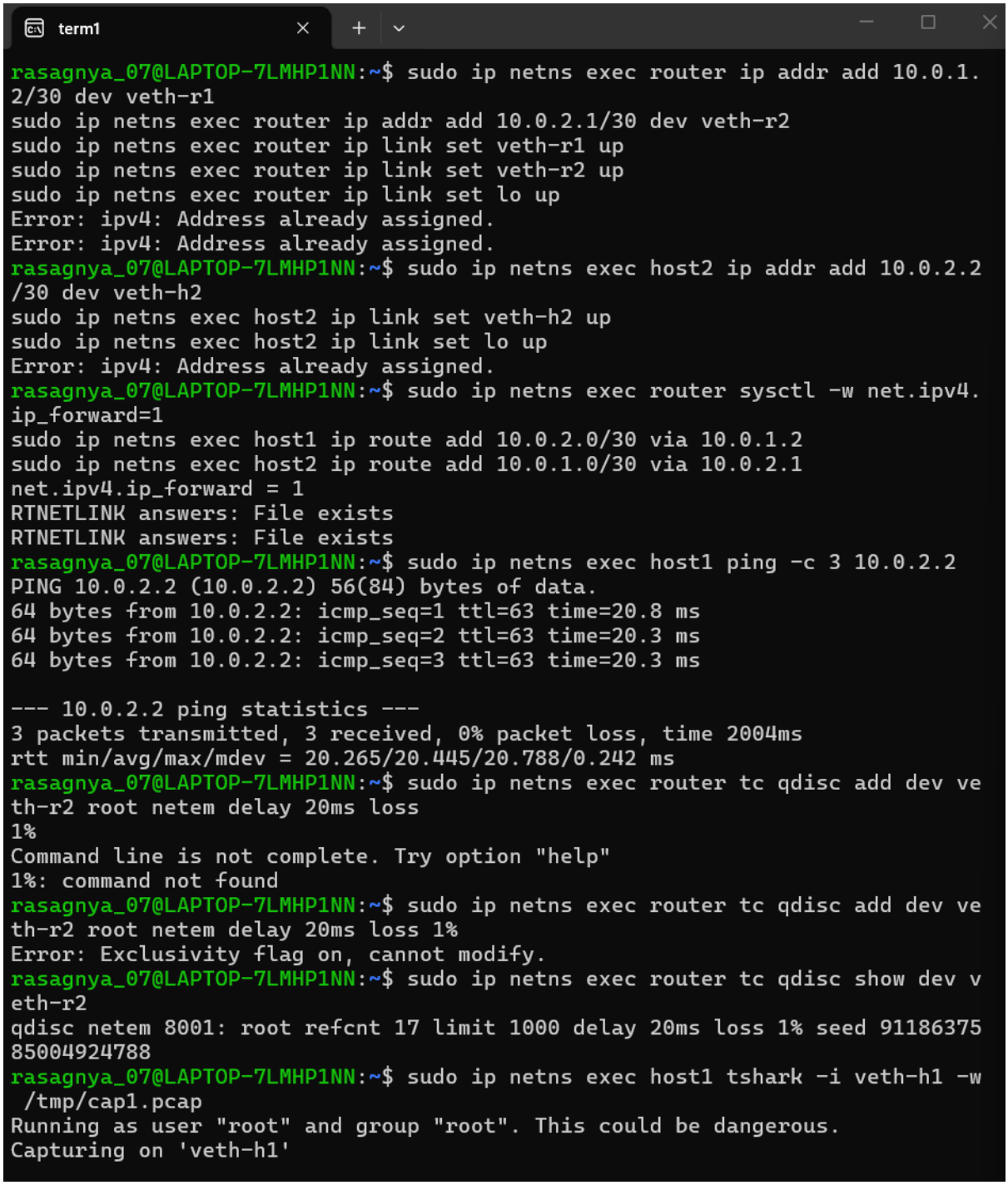
```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=20.8 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=20.3 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=20.3 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 20.265/20.445/20.788/0.242 ms
```

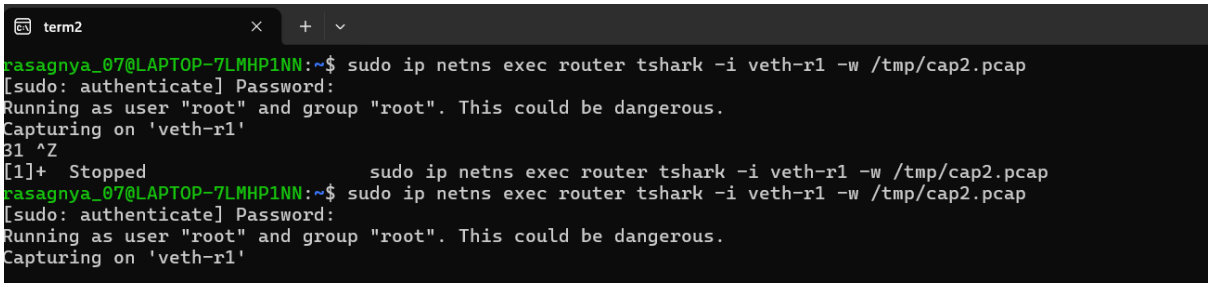
Add delay and loss Verify:

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec router tc qdisc add dev ve
th-r2 root netem delay 20ms loss
1%
Command line is not complete. Try option "help"
1%: command not found
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec router tc qdisc add dev ve
th-r2 root netem delay 20ms loss 1%
Error: Exclusivity flag on, cannot modify.
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec router tc qdisc show dev v
eth-r2
qdisc netem 8001: root refcnt 17 limit 1000 delay 20ms loss 1% seed 91186375
85004924788
```

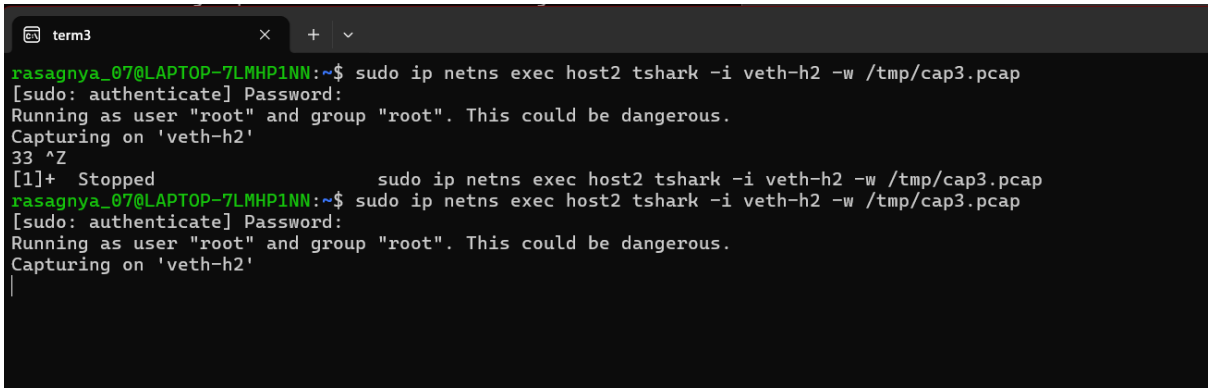
TERMINAL 1 — Capture at host1



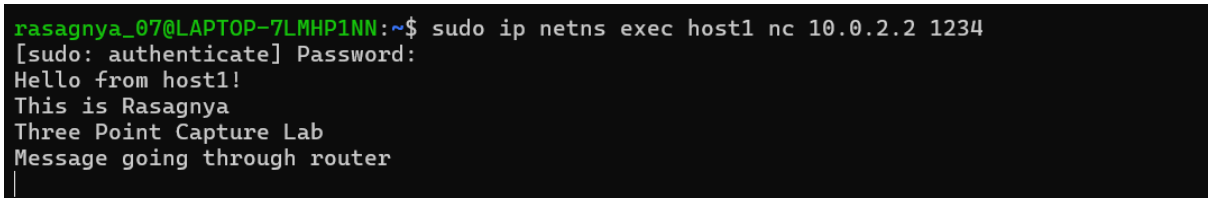
TERMINAL 2 — Capture at router

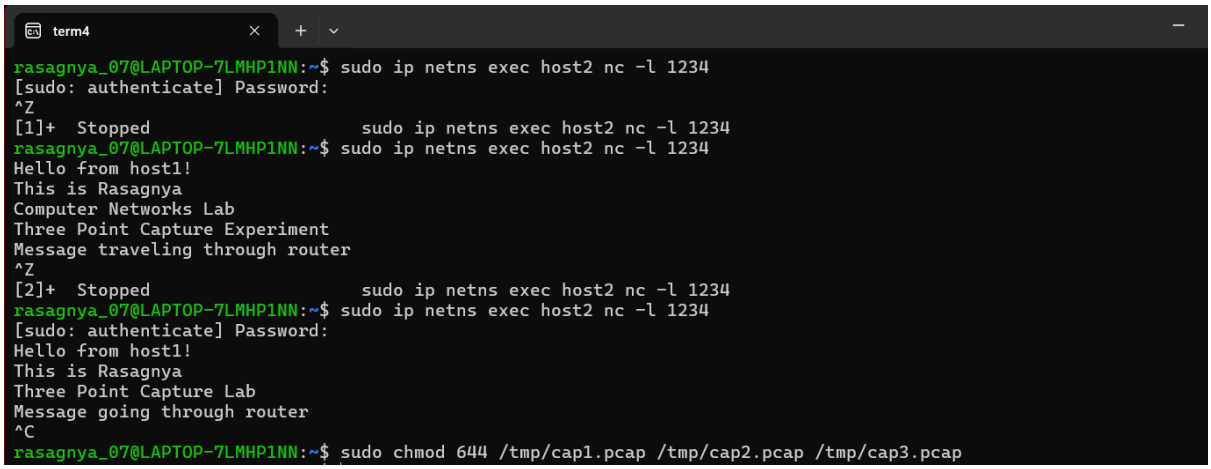


TERMINAL 3 — Capture at host2



TERMINAL 4 — Send the message





Check files are not empty:

```
term4
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:
^Z
[1]+  Stopped                 sudo ip netns exec host2 nc -l 1234
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host2 nc -l 1234
Hello from host1!
This is Rasagnya
Computer Networks Lab
Three Point Capture Experiment
Message traveling through router
^Z
[2]+  Stopped                 sudo ip netns exec host2 nc -l 1234
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:
Hello from host1!
This is Rasagnya
Three Point Capture Lab
Message going through router
^C
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
rasagnya_07@LAPTOP-7LMHP1NN:~$ ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 2.6K Nov 13 03:47 /tmp/cap1.pcap
-rw-r--r-- 1 root root 2.6K Nov 13 03:47 /tmp/cap2.pcap
-rw-r--r-- 1 root root 2.6K Nov 13 03:47 /tmp/cap3.pcap
```

Check TTL at all 3 points:

```
term4
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host2 nc -l 1234
Hello from host1!
This is Rasagnya
Computer Networks Lab
Three Point Capture Experiment
Message traveling through router
^Z
[2]+  Stopped                 sudo ip netns exec host2 nc -l 1234
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:
Hello from host1!
This is Rasagnya
Three Point Capture Lab
Message going through router
^C
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
rasagnya_07@LAPTOP-7LMHP1NN:~$ ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 2.6K Nov 13 03:47 /tmp/cap1.pcap
-rw-r--r-- 1 root root 2.6K Nov 13 03:47 /tmp/cap2.pcap
-rw-r--r-- 1 root root 2.6K Nov 13 03:47 /tmp/cap3.pcap
rasagnya_07@LAPTOP-7LMHP1NN:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      7 63
      8 64
      7 63
      8 64
      8 63
      7 64
```

Read your actual message from router capture:

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e tcp.payload
2>/dev/null | xxd | head -20



48656c6c6f2066726f6d20686f737431210a

5468697320697320526173617361676e79610a

546872656520506f696e7420436170747475726520c61620a

4d65737361676520676f696e67207468726f75676820726f757465720a
```

Check delay:

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ tshark -r /tmp/cap1.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T
fields -e frame.time_relative 2>/dev/null
tshark -r /tmp/cap2.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T
fields -e frame.time_relative 2>/dev/null
tshark -r /tmp/cap3.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T
fields -e frame.time_relative 2>/dev/null
tshark: option requires an argument: T

Usage: tshark [options] ...

Capture interface:
  -i <interface>, --interface <interface>
                           name or idx of interface (def: first non-loopback)
  -f <capture filter>      packet filter in libpcap filter syntax
  -s <snaplen>, --snapshot-length <snaplen>
                           packet snapshot length (def: appropriate maximum)
  -p, --no-promiscuous-mode
                           don't capture in promiscuous mode
  -I, --monitor-mode       capture in monitor mode, if available
  -B <buffer size>, --buffer-size <buffer size>
                           size of kernel buffer in MiB (def: 2MiB)
  -y <link type>, --linktype <link type>
                           link layer type (def: first appropriate)
  --time-stamp-type <type> timestamp method for interface
  -D, --list-interfaces    print list of interfaces and exit
  -L, --list-data-link-types
                           print list of link-layer types of iface and exit
  --list-time-stamp-types  print list of timestamp types for iface and exit
```

Check retransmissions:

```
rasagnya_07@LAPTOP-7LMHP1NN:~$ tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
0
0
```

Check RTT:

```
term4

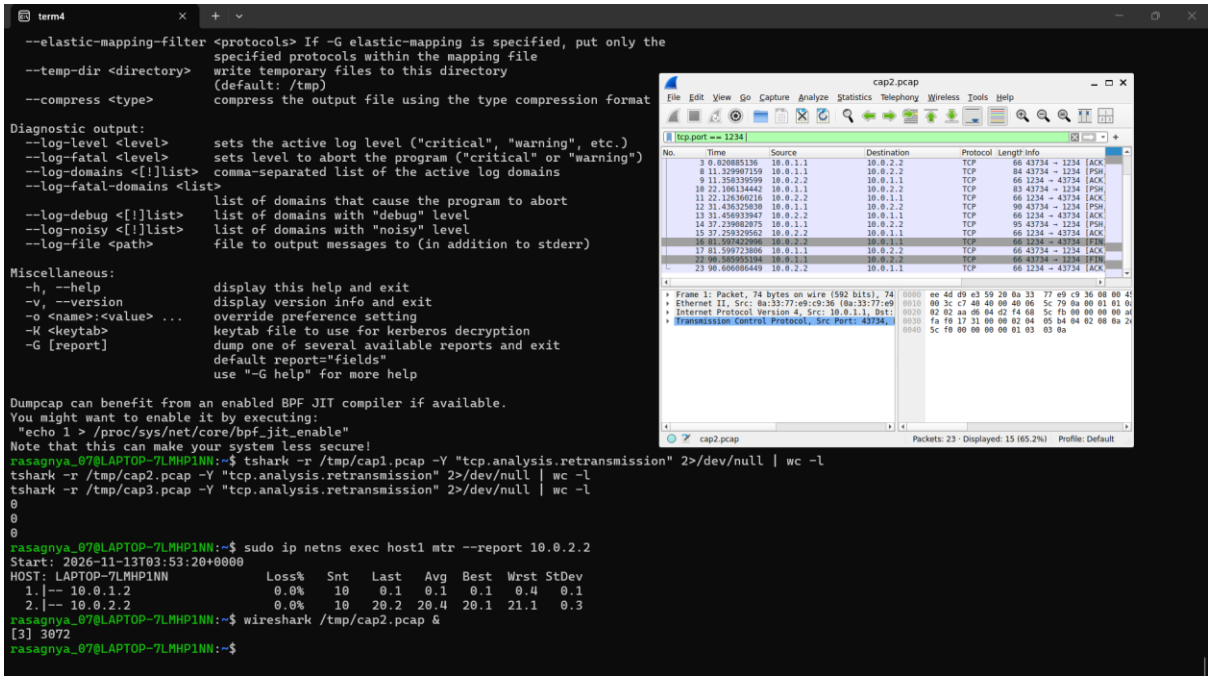
  --elastic-mapping-filter <protocols> If -G elastic-mapping is specified, put only the
                           specified protocols within the mapping file
  --temp-dir <directory>   write temporary files to this directory
                           (default: /tmp)
  --compress <type>        compress the output file using the type compression format

Diagnostic output:
  --log-level <level>      sets the active log level ("critical", "warning", etc.)
  --log-fatal <level>      sets level to abort the program ("critical" or "warning")
  --log-domains <[!]list>  comma-separated list of the active log domains
  --log-fatal-domains <list>
                           list of domains that cause the program to abort
  --log-debug <[!]list>    list of domains with "debug" level
  --log-noisy <[!]list>    list of domains with "noisy" level
  --log-file <path>        file to output messages to (in addition to stderr)

Miscellaneous:
  -h, --help               display this help and exit
  -v, --version            display version info and exit
  -o <name>:<value> ...    override preference setting
  -K <keytab>              keytab file to use for kerberos decryption
  -G [report]              dump one of several available reports and exit
                           default report="fields"
                           use "-G help" for more help

Dumpcap can benefit from an enabled BPF JIT compiler if available.
You might want to enable it by executing:
 "echo 1 > /proc/sys/net/core/bpf_jit_enable"
Note that this can make your system less secure!
rasagnya_07@LAPTOP-7LMHP1NN:~$ tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
0
0
rasagnya_07@LAPTOP-7LMHP1NN:~$ sudo ip netns exec host1 mtr --report 10.0.2.2
Start: 2026-11-13T03:53:20+0000
HOST: LAPTOP-7LMHP1NN             Loss%   Snt   Last   Avg  Best  Wrst StDev
  1.|-- 10.0.1.2                   0.0%    10    0.1   0.1   0.1   0.4   0.1
  2.|-- 10.0.2.2                   0.0%    10   20.2  20.4  20.1  21.1   0.3
```

TERMINAL 4 — Open Wireshark



Filter your message only:

